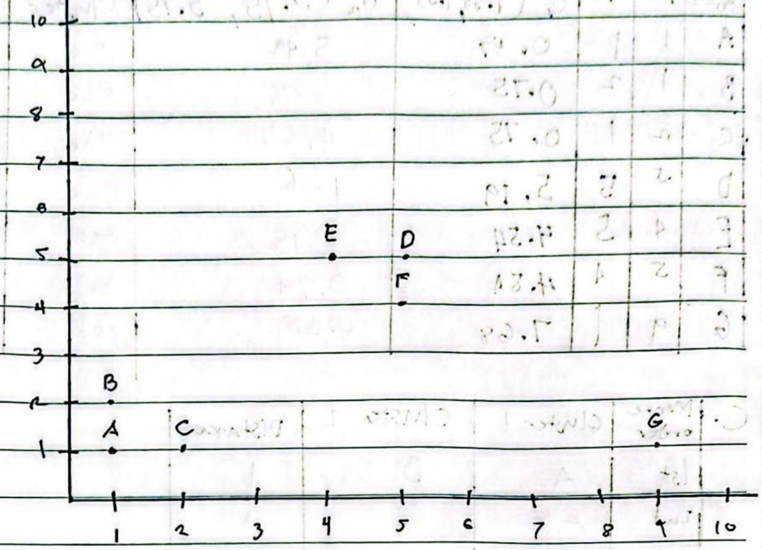
NO.:
DATE:

Custer Jeremiah P. Valencerina COM232

A.

Data Points	X	Y
A	1	1
B	1	2
C	2	1
D	5	5
E	4	5
F	5	4
G	9	1



	A (1,1)	B (1,2)	C (2,1)	D (5,5)	E (4,5)	F (5,4)	G (9,1)
A (1,1)	0	1	1	5.66	5	5	8
B (1,2)	1	0	1.41	5	4.24	4.47	8.06
C (2,1)	1	1.41	0	5	4.47	4.24	7
D (5,5)	5.66	5	5	0	1	1	5.66
E (4,5)	5	4.24	4.47	1	0	1.41	6.40
F (5,4)	5	4.47	4.24	1	1.41	0	5
G (9,1)	8	8.06	7	5.66	6.40	5	0

B.

Data Points	X	Y	d_1 (1,1)	d_2 (5,5)	Cluster
A	1	1	0	5.66	Blue
B	1	2	1	5	Blue
C	2	1	1	5	Blue
D	5	5	5.66	0	Red
E	4	5	5	1	Red
F	5	4	5	1	Red
G	9	1	8	5.66	Red

New d_1

$a_1 = \frac{1+1+2}{3} = 1.33$

$b_1 = \frac{1+2+1}{3} = 1.33$

New d_2

$a_2 = \frac{5+4+5+9}{4} = 5.75$

$b_2 = \frac{5+5+4+1}{4} = 3.75$

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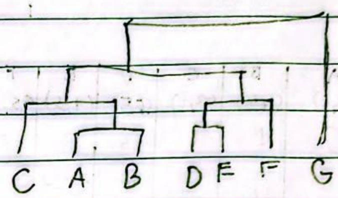
Data Points	X	Y	Blue d_1 (1.33, 1.33)	Red d_2 (5.75, 3.75)	Cluster
A	1	1	0.47	5.49	Blue
B	1	2	0.75	5.06	Blue
C	2	1	0.75	4.65	Blue
D	5	5	5.19	1.46	Red
E	4	5	4.54	2.15	Red
F	5	4	4.51	0.79	red
G	9	1	7.68	4.26	red

C.

Merge order	Cluster 1	Cluster 2	Distance
1st	A	B	1
2nd	AB	C	1
3rd	D	E	1
4th	DE	F	1
5th	CAB	DEF	4.24
6th	CABDEF	G	5



D. eps = 1, min_samples = 2

Data Point	Neighbors	No. of Neighbors	Point Type	Cluster
A	B, C	2	Core	Blue
B	A	1	Non core	Blue
C	A	1	Non core	Blue
D	E, F	2	Core	Red
E	D	1	Non core	Red
F	D	1	Non core	red
G	None	0	Noise	Outlier